

Ana-Catalina Villalobos Contardo

DATA SCIENTIST & ML ENGINEER

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SUMMARY

Civil Engineer with a solid background in Data Science, Machine Learning (ML), Data Engineering, and Backend Development. Combines a strong theoretical foundation (mathematics, statistics) with practical experience in developing and implementing ML solutions (predictive maintenance, operational optimization). Specialist in designing robust data pipelines (ETL/ELT), interpretable predictive models, and complex integration architectures. Proven technical leadership managing multidisciplinary teams, with a clear focus on solving complex problems, generating strategic value, and operational excellence.

EXPERIENCE

SimpliRoute | *Learning Engineer* | August 2025 - Present

- Design and implementation of ML solutions for route optimization and geolocation, focusing on georeferencing accuracy.
- Management and orchestration of large-scale data flows using Airflow and Airbyte, ensuring process quality, traceability, and reliability.
- Collaboration with Product, Engineering, and Operations teams to validate and communicate technical findings.

Fracttal | *Tech Lead Fracttal Hub* | November 2023 - July 2025

- Led a multidisciplinary integration team, achieving product development goals and client projects, while prioritizing team mentoring.
- Strategic planning of the product roadmap, ensuring alignment with business objectives and continuous value delivery.
- Python development for data processing and orchestration libraries, enabling integrations with multiple systems.
- Supervision of integration projects, ensuring deadlines and quality in data flows across heterogeneous systems.

Fracttal | *Data Scientist* | August 2021 - November 2023

- Designed and implemented data science processes for predictive maintenance, asset prioritization, and failure estimation.
- Collaborated on product creation and development to manage integrations between Fracttal and other platforms.

Fracttal | *Data Analyst* | February 2020 - July 2021

- Developed a predictive maintenance tool using advanced data analysis, predictive statistics, and Machine Learning techniques.
- Database design and management alongside web development tasks.

TOOLS & TECHNICAL SKILLS

- **Programming Languages:** Python (Principal/Advanced) | SQL | JavaScript | TypeScript | MATLAB | Java.
- **Cloud & DevOps:** Google Cloud Platform (BigQuery, Cloud Storage, Looker Studio, Google Pub/Sub) | Git | GitHub | Bitbucket.
- **Frameworks & Libraries:** Pandas & NumPy | Scikit-learn | LangChain | TensorFlow | Django | React JS.
- **Databases:** PostgreSQL | SQL Server | MySQL | MongoDB.
- **Containers & Orchestration:** Airflow | Airbyte | Docker | Kubernetes.
- **Management & Communication:** Jira | MS Planner | MS Teams | Sharepoint | Notion | Slack | Trello.
- **Key Skills:** Data Extraction & Processing | Data Visualization | Relational DB Design | Technical Leadership | Agentic AI.

EDUCATION

- **Universidad de Chile** | *Civil Engineering with a minor in structures, construction, and geotechnics* | Graduated 2019, Degree awarded 2020 (Maximum Distinction)
Thesis: Gaussian Processes and vibration modes for delamination detection in composite material plates.
- **Universidad de Chile** | *Bachelor of Engineering Sciences, Civil minor* | Completed 2017
Notable activity: Teaching Assistant for Probabilities and Statistics.

PUBLICATIONS

- **Generalized Gaussian smoothing for baseline-free debonding assessment of sandwich panels** | *Structural Control Health Monitoring* - March 2021 (<https://doi.org/10.1002/stc.2727>).
Methodological proposal for delamination detection and evaluation in composite plates using Gaussian Processes and Bayesian analysis.

LANGUAGES & CERTIFICATIONS

- **Languages:** Spanish (Native) | English (Advanced - EFSET English Certificate C2 Proficient, 2020)
- **Certifications:** React & Django Full Stack (Udemy, 2022) | Mastering Data Visualization (Udemy, 2022) | Applied Machine Learning in Python (Coursera, 2020).